

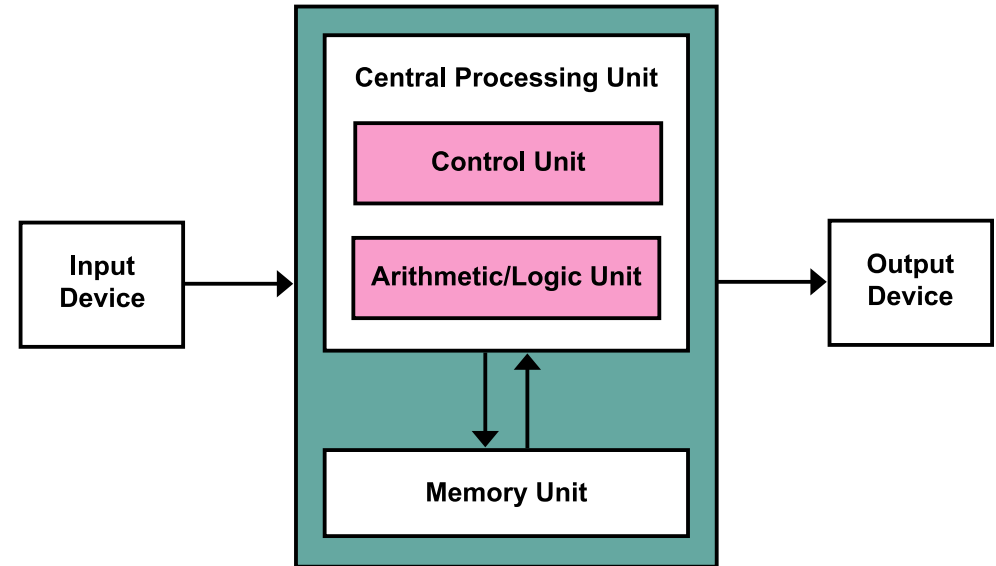
GUIN-16

An Original 16-Bit CPU

Jacob Shogren

<https://github.com/shogrenjacob/16-bit-cpu>

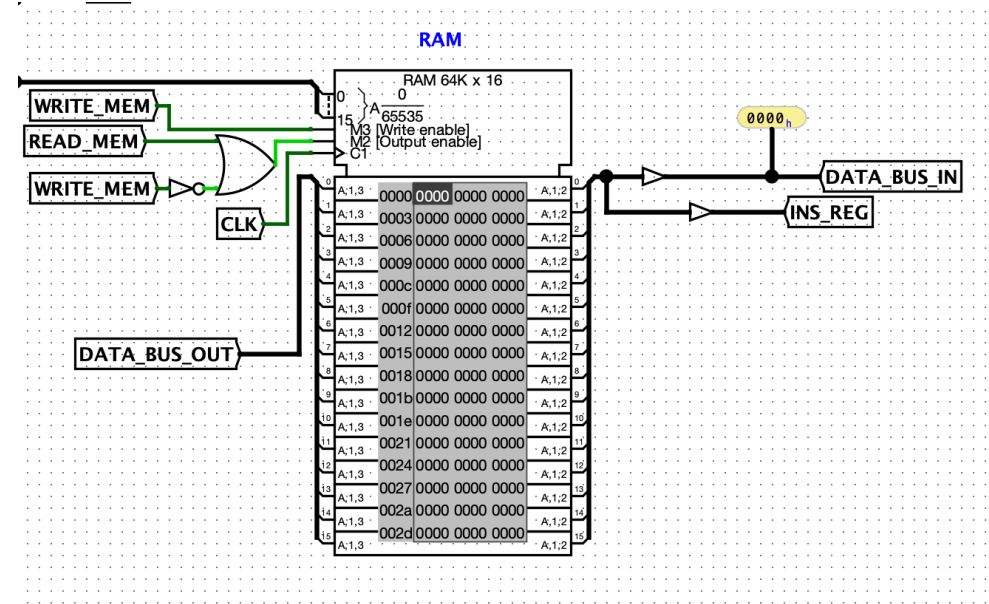
- Von Neumann Architecture
- 65,535 memory addresses (255 per page)
- 7 Registers, 5 Flags
- Two's Complement
- Microcode Instruction Set Implementation



By Kapooht - Own work, CC BY-SA 3.0,

<https://commons.wikimedia.org/w/index.php?curid=25789639>

- Stack Memory (0x00 → 0xfe)
- Free Memory (0xff → 0xfffc)
- Reset Vector (0xfffd → 0xffff)

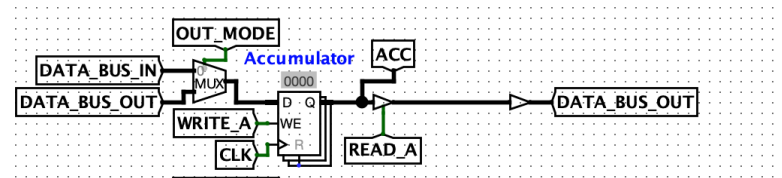
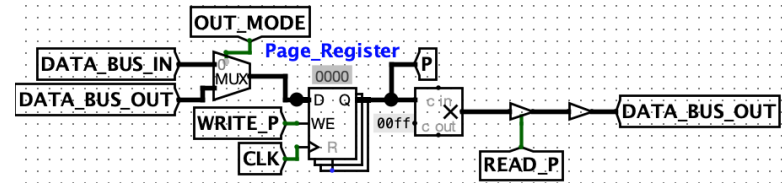
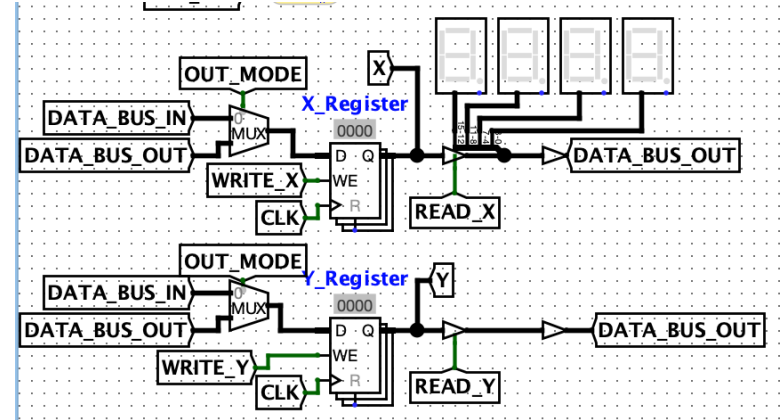


0x00 - 0xFE (Stack Memory)

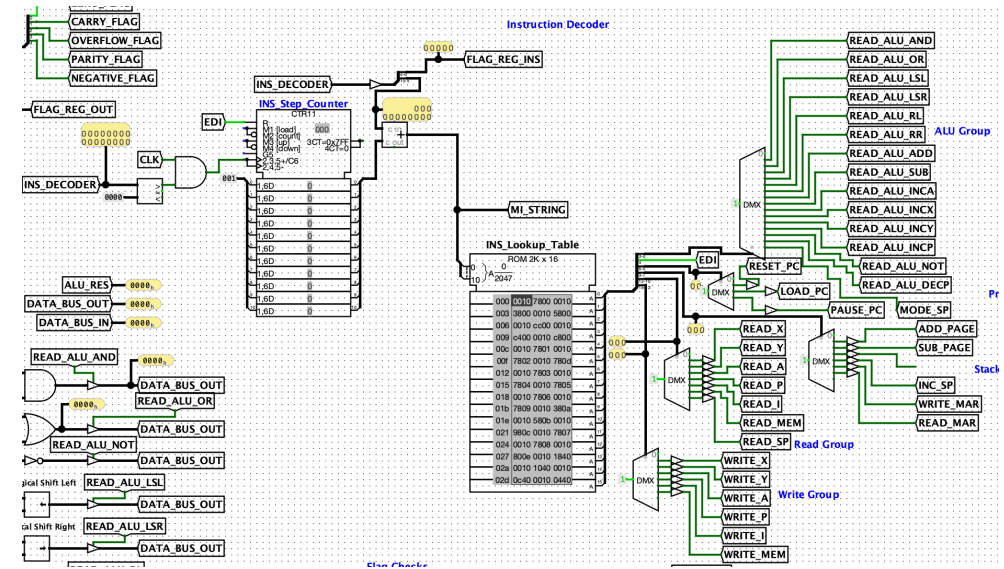
0xFF - 0xFFFFC (Program Memory)

0xFFFFD - 0xFFFF (Reset Vector / EOM)

- General-Purpose Registers
- Accumulator
- Instruction Register
- Page Register
- Flag Register
- Memory Address Register (MAR)



- Takes the instruction stored in the Instruction Register and breaks it into parts.
- Instruction = 11 most significant bits
- Flag Checks = 5 least significant bits
- Instruction part loaded into the Instruction Lookup Table



Six groups of microcode:

Write
011

Read
110

Indirect
100

PC
00

EDI
0

ALU
1111

Concatenate these together to get a microinstruction:

0111101000001111 = 0x7a0f

Writing to Accumulator, reading from memory, moving the Stack Pointer, and changing the Stack Pointer mode all in one clock tick!

- Live in memory and are sent to the Instruction Register to be processed in the Control Unit.
- Made up of a series of microinstructions
- Instruction broken into three parts:
 - Hex Digit 1 = Addressing Mode
 - Hex Digit 2 & 3 = Lookup Table Index
 - Hex Digit 4 = Flags to Compare

LDX Immediate (0x0020)

STA PageAhead (0x5200)

ADC Absolute (0x453)

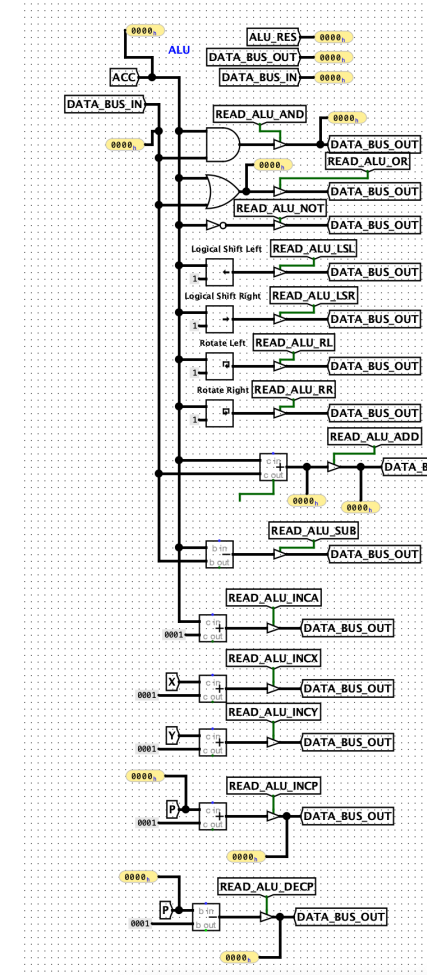
JMP Indirect (0x7660)

PSHY Implied (0x0700)

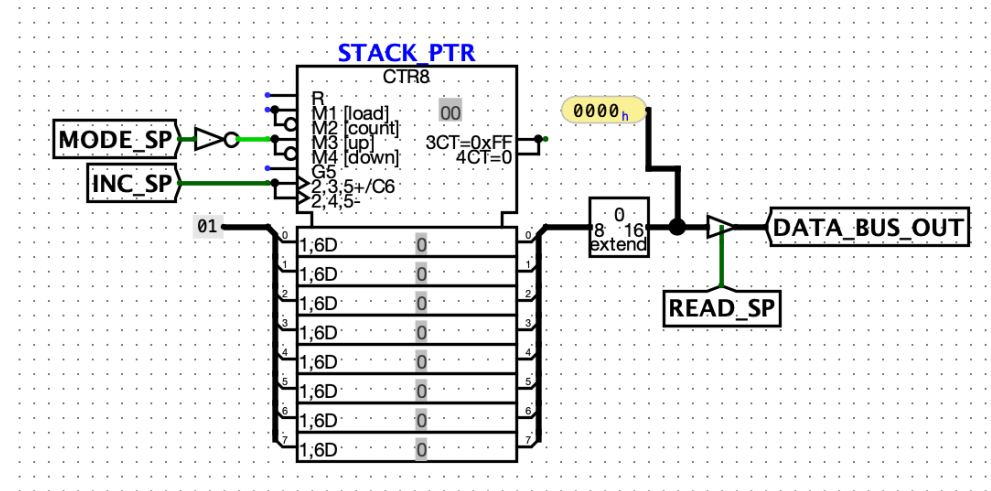
There are 8 addressing modes in the Guin-16:

1. **Implicit** - No data needed for this instruction.
2. **Immediate** - Next address consecutive address in memory holds the data.
3. **Page Ahead** - Data found at the address a page ahead of instruction.
4. **Page Behind** - Data found at the address a page behind the instruction.
5. **Absolute** - Data at next address points to the address of the data to use.
6. **AbsoluteX** - Add the contents of X to absolute address found.
7. **AbsoluteY** - Add the contents of Y to absolute address found.
8. **Indirect** - Perform two absolutes to get the data needed.

- Handles arithmetic and logical operations in the CPU
- Uses Carry-Lookahead Addition
- Two-operand operations *always* use Accumulator
- Intermediate operations sent to Accumulator



- Points to the current top of the Stack
- Increments every push operation
- Decrements every pop operation



Documentation & Demonstration

